

Jaeyun Sim

Ph.D. Candidate in Mechanical Engineering — Robotics and Force Control

Sungkyunkwan University, Republic of Korea

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Research Profile

Ph.D. candidate developing learning-based adaptive force-control methods for robotic contact tasks in uncertain and dynamic environments. Research spans reinforcement learning, variable admittance control, nonlinear and passivity-based control, force/torque sensing, and signal processing. Experienced in real-time implementation and experimental validation on collaborative manipulators. Seeking postdoctoral research opportunities in robot learning, interaction control, and intelligent manufacturing.

Research Interests

Robot learning for contact-rich manipulation; adaptive force and impedance/admittance control; safe physical human-robot interaction; reinforcement and imitation learning; robotic finishing and intelligent manufacturing; force/torque sensing and calibration.

Education

Ph.D. in Mechanical Engineering

Sungkyunkwan University, Republic of Korea

Expected Feb. 2027

- Research focus: *G-Force: Generalized Force Control via Environment-Encoder-Driven Reinforcement Learning-Based Variable Admittance Control.*

M.S. in Mechanical Engineering

Gwangju Institute of Science and Technology (GIST), Republic of Korea

Aug. 2020

- Thesis: *Tool Path Generation of a Lapping Wheel Considering Wheel Compliance.*

B.S. in Mechanical Engineering

Chungnam National University, Republic of Korea

Feb. 2019

Research Experience

Visiting Researcher

Manipulation and Mobility Systems Group, National Institute of Standards and Technology (NIST), USA

Jun.–Oct. 2025

- Investigated reinforcement-learning-based force control for stable adaptation in uncertain environments.
- Contributed to performance-evaluation methods for force control across environments with varying stiffness.

Graduate Research Assistant

Robotics Innovatory, Sungkyunkwan University, Republic of Korea

Dec. 2020–Present

- Developing generalized force control using environment-encoder-driven reinforcement learning and variable admittance control.
- Developed dynamical-system-based fuzzy adaptive admittance control with passivity management using an energy tank; implemented on a UR10 manipulator using ROS, C++, Linux, and PREEMPT_RT.
- Developed multi-channel weighted band-stop filtering and energy-tank-based adaptive admittance control; validated on an Indy7 manipulator using ROS, C++, Linux, and Xenomai.
- Developed a calibration control system for an ultra-thin torque sensor, combining direct force control, trajectory generation, and neural-network-based calibration.

Graduate Researcher

Ultraprecision Machine System Laboratory, GIST, Republic of Korea

Mar. 2019–Oct. 2020

- Generated optimal tool paths that prevent over-machining using an identified compliant lapping-wheel model.
- Designed control methods for a gantry-type five-axis manipulator for automated surface lapping.

Undergraduate Research Assistant

Manufacturing System Control Laboratory, Chungnam National University, Republic of Korea

- Planned tool paths and designed and controlled a five-axis manipulator for post-processing additively manufactured parts.

Research Intern

Ultra Precision Equipment Laboratory, Korea Institute of Machinery and Materials (KIMM), Republic of Korea

Jan.–Feb. 2018

- Designed and controlled a tilting head for a mobile machine tool.

Peer-Reviewed Journal Publications

1. **Sim, J.**, Kim, Y., Kim, E. C., Song, E., Lee, S., Sim, J., and Choi, H. R. (2026). "Dynamical System-Based Fuzzy Adaptive Admittance Control for Uncertain Environments." *Electronics*, 15(10), 2045.
2. Kim, E. C., **Sim, J.**, Song, E., Lee, S. Y., Sim, J. Y., and Choi, H. R. (2026). "Hand-Gesture Based Robot Tracking and Teaching Framework: Demonstration in Surface Polishing Task." *IEEE Robotics and Automation Letters*.
3. Um, S., Son, Y. G., **Shim, J.**, and Choi, H. R. (2026). "Toward Reliable Bin-Picking: Collision-Aware Robotic Design and Control Strategy for Heavily Cluttered Environment." *IEEE Robotics and Automation Practice*.
4. Lee, S. Y., Sim, J. Y., Seok, D. Y., Kim, Y. B., **Shim, J.**, Kim, U., and Choi, H. R. (2026). "A Redundant-Sensing-Based Six-Axis Force/Torque Sensor Enabling Compactness and High Sensitivity." *Sensors*, 26(3), 871.
5. Moon, S. J., Yim, H., Kang, H., Shim, J. Y., Jung, D., and Choi, H. R. (2025). "Safety-Critical Reactive Motion Using Constrained Variable Admittance Control With Dual-Type Proximity Sensors." *IEEE Robotics and Automation Letters*.
6. Lee, S. Y., Sim, J. Y., Kim, Y. B., Seok, D., **Shim, J.**, and Choi, H. R. (2025). "Design Analysis and Isotropic Optimization for Miniature Capacitive Force/Torque Sensor." *Sensors*, 25(3), 940.
7. Park, S., Koh, D., **Shim, J.**, Kim, J. J., and Lee, S. K. (2021). "Gantry Type Lapping Manipulator Toward Unmanned Lapping Process for a Large Work Surface." *International Journal of Precision Engineering and Manufacturing-Green Technology*, 8, 1723–1737.
8. Song, J., **Shim, J.**, Kim, N., Chae, G. B., and Lee, W. (2019). "Friction Compensation Control of Feed Drive System Operated in Vacuum." *Journal of Mechanical Science and Technology*, 33(11), 5353–5360.
9. **Shim, J.**, Kim, N., Oh, D., Hong, S., and Lee, W. (2019). "Design of a Robot Machining System and Tool Path Generation for Post-Process Machining of 3D Printed Product." *Transactions of the Korean Society of Mechanical Engineers A*, 43(5), 359–364.

Conference Presentations

1. **Shim, J.**, Park, S. K., Koh, D. G., Kim, J., and Lee, S. K. "Design and Control of a 5-Axis Manipulator for the Automatic Lapping Process." ASPEN 2019, Matsue, Japan, 2019. Oral presentation.
2. **Sim, J.**, Song, E. S., and Choi, H. R. "Test and Evaluation Standard for Compliance Control Performance to Ensure the Contact Safety and Performance of Field-Deployable Robotic Systems." Society for Standards, Certification and Safety, 2026. Oral presentation; Excellent Paper Award.
3. **Shim, J.** and Lee, S. K. "Investigation of Optimum Machining Conditions Considering Compliance of Lapping Wheel." Korean Society for Precision Engineering Combined Conference, 2020. Oral presentation; Best Paper Award.
4. **Shim, J.** and Lee, S. K. "Automatic Lapping System Tool Path Generation Using Direct Scanning and Bi-Cubic Spline." Korean Society for Precision Engineering Autumn Conference, Changwon, 2019. Poster.
5. **Shim, J.**, Park, S. K., Koh, D. G., and Lee, S. K. "Analysis of a 5-Axis Manipulator for the Finishing Application Considering Robot Dynamic Model." Korean Society for Precision Engineering Spring Conference, Jeju, 2019. Poster.
6. **Shim, J.**, Hong, S., Seo, M. S., Kwak, J., and Lee, W. "Robot Machine Tool Based Post-Processing for 3D Printing." Korean Society of Manufacturing Technology Engineers Spring Conference, Daejeon, 2018. Poster; Excellent Paper Award.
7. **Shim, J.**, Kim, N., Kim, C. J., Ro, S.-K., and Lee, W. "Control Technique for Tilting Head of Mobile Machine Tool." Korean Society of Manufacturing Technology Engineers Spring Conference Special Section, Daejeon, 2018. Oral presentation.
8. **Shim, J.**, Kim, N., Hong, S., Seo, M. S., Kwak, J., and Lee, W. "5-DOF Robotic Machining System for Complex Shape Machining." Korean Society for Precision Engineering Autumn Conference, Gwangju, 2017. Poster.
9. Song, J., **Shim, J.**, Kim, N., Chae, G. B., and Lee, W. "Friction Characteristic of Feed Drive System Operated in Vacuum." Korean Society for Precision Engineering Spring Conference, Jeju, 2017. Poster.

Awards and Honors

- 2026 Excellent Paper Award, Society for Standards, Certification and Safety
- 2020 Best Paper Award, Korean Society for Precision Engineering
- 2018 Excellent Paper Award, Korean Society of Manufacturing Technology Engineers
- 2018 Excellent Award, Industry-Linked X-Corps Contest
- 2018 Excellent Prize, CNU Engineering Fair Creative Engineering Capstone Design and Graduation Competition

Selected Technology Demonstrations

- **Robot World 2024:** Integrated vision, remote tracking, and force control for automated polishing of complex surfaces using AIDIN Robotics technology.
- **Smart Factory + Automation World 2024:** Physical-HRI hand-guiding and force-controlled automated polishing of complex surfaces.
- **Robot World 2023:** Physical-HRI hand-guiding and force-controlled automated polishing of flat surfaces.
- **Robot World 2022:** Stability-enhanced admittance-control hand-guiding using dual six-axis force/torque sensors and an Indy7 collaborative robot.
- **Robot World 2021:** Admittance-control hand-guiding and collision detection using an ultra-thin joint torque sensor.

Technical Skills

Programming	C++, Python, MATLAB; ROS/ROS 2
Simulation	Simulink, Gazebo, NVIDIA Isaac Sim
Control	Reinforcement-learning-based force control; admittance/impedance control; passivity-based control; fuzzy control; model predictive control
Learning	Reinforcement learning; force-aware imitation learning; artificial neural networks for force/torque sensor calibration
Signal Processing	Fast Fourier transform; frequency-selective filtering; Kalman filtering
Systems	Linux; PREEMPT_RT; Xenomai; real-time robotic control
Hardware	UR10/UR10e and Indy7 manipulators; six-axis force/torque sensors; torque sensors; IMUs

Professional Affiliation

Student Member, IEEE Robotics and Automation Society.